University Library Management System

Assignment 1

Project planning and GitHub

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Software Engineering | Assignment 1 | Prepared 12 November 2026

Project planning window: 13 November–14 December 2026 (proposed). This report covers Assignment 1. Later assignments appear in the WBS and schedule as planned work.

112 base person-hours · 16 work packages · 12 risks
One named student; three functional roles

Repository: https://github.com/talalahmad-ai/university-library-management-system.

Prepared for Talal Ahmad with AI assistance. Repository technical work used the authorized collaborator Hamza-Khaalid; actual commit authors and timestamps are preserved.

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This report presents the project definition, WBS, schedule, responsibility/effort allocation and risk register. The accompanying ZIP contains the editable workbook and planning sources. The appendices include genuine GitHub evidence supporting the repository, README, commits and assigned Issues.

1. Project definition and scope

The case scenario describes a university library using manual or outdated records for borrowing, returns, membership and overdue books. The academic project analyzes and designs an LMS; complete implementation is not required. Phase 1 produces planning and GitHub evidence.

Capability	Planned analysis/design boundary
Catalog and copies	Searchable books and copy availability; no live inventory import.
Membership	Student/faculty membership and librarian/admin roles; policies unconfirmed.
Borrowing and returns	Member validation, available-copy loans, due dates, return and overdue handling.
Reservations	Search/detail/reserve interaction; queue precedence and expiry need confirmation.
Overdue notifications	Notification information and triggers as design concepts; delivery rules unconfirmed.

Excluded: production backend/frontend, live messaging, payment processing, SSO, hosting, real record migration and fabricated human evidence. Success is a coherent, traceable and reviewable coursework package. See reports/project-charter.md for full assumptions, governance and acceptance measures. Team member: Talal Ahmad (@talalahmad-ai). The work plan allocates all three functional roles to this one student. Hamza-Khaalid is the authorized technical helper; preparation commits retain Codex attribution. Dates and daily capacity are explicit planning assumptions.

Objectives

Define the library system boundary and intended benefits; establish a controlled repository; plan requirements, modeling and UI work; allocate responsibilities and effort; assess risks; and maintain verifiable commit and task evidence.

Assumptions and constraints

Item	Planning basis
Team	One confirmed student, Talal Ahmad, holds Lead, Analyst and Designer roles.
Dates	13 November–14 December 2026 is a proposed calendar-day planning window; official deadlines were not supplied.
Capacity	Six hours per calendar day across all roles is a planning estimate, including weekends.
Library policy	Loan periods, quotas, penalties and reservation precedence require stakeholder clarification in the requirements phase.
Scope	Analysis, planning, UML and a clickable UI are planned; a complete working LMS is not required.
Information	Use fictional demonstration data and genuine repository evidence; preserve actual authors and timestamps.

2. Work Breakdown Structure

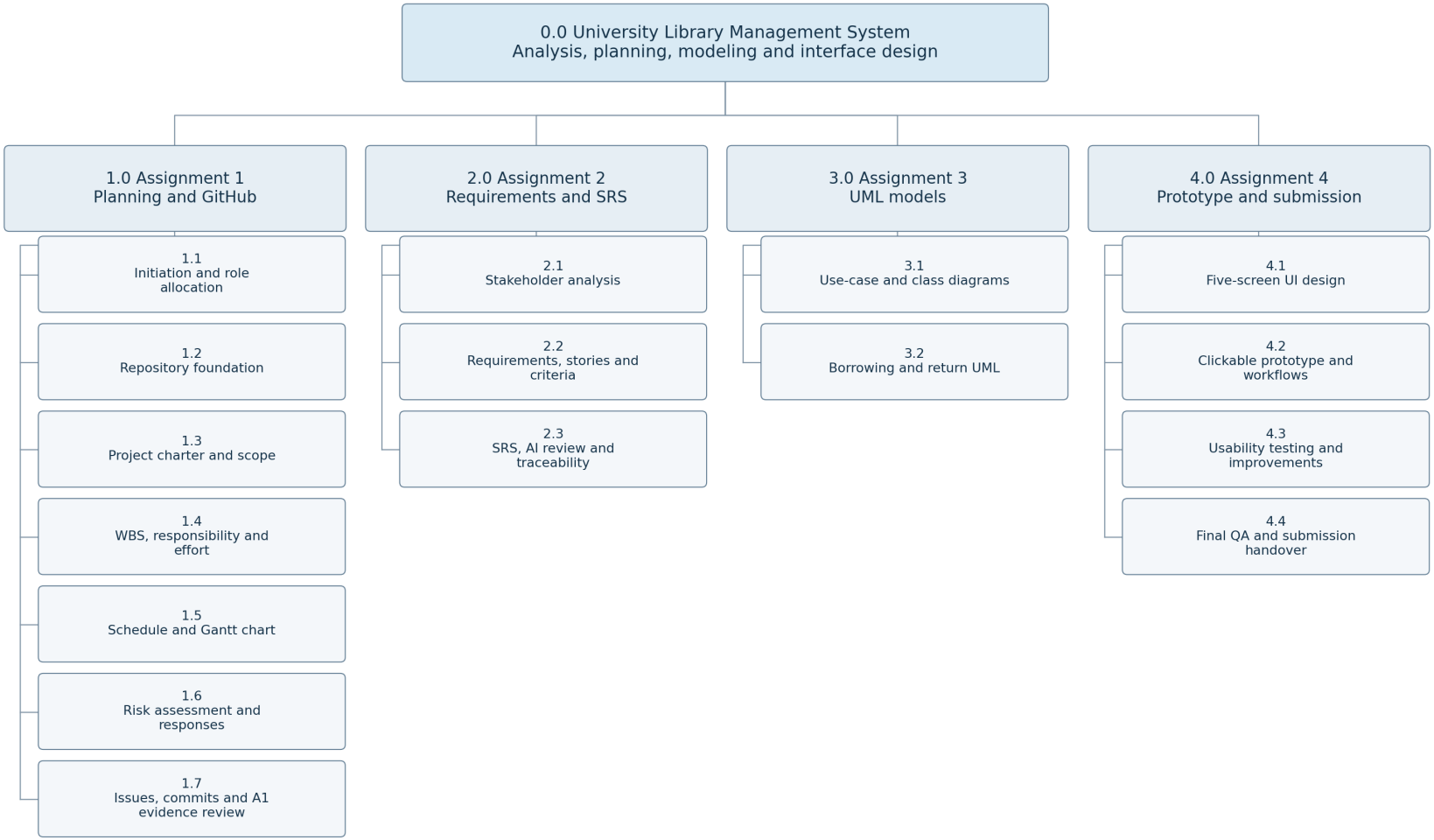
Root 0.0 contains four assignment branches. The following 16 leaf packages are effort-bearing; summary branches and milestone gates add no duplicate effort. Editable diagram: reports/wbs.drawio.

ID	Package	Deliverable
1.1	Initiation and role allocation	Source review, named student and role allocation
1.2	Repository foundation	README, four tracked folders and preserved remote history
1.3	Project charter and scope	Charter with objectives, boundaries, assumptions and success criteria
1.4	WBS, responsibility and effort	WBS source/export, named RACI and task/person effort totals
1.5	Schedule and Gantt chart	Editable schedule and dated Gantt with milestones
1.6	Risk assessment and responses	Scored risk register with triggers, mitigations and contingencies
1.7	Issues, commits and A1 evidence review	Assigned Issues, genuine remote commits, screenshots and A1 audit
2.1	Stakeholder analysis	Stakeholder-needs table and policy questions
2.2	Requirements, stories and criteria	Requirements list, user stories and measurable acceptance criteria
2.3	SRS, AI review and traceability	Structured SRS, AI-review decisions and traceability matrix
3.1	Use-case and class diagrams	Editable use-case/class diagrams, exports and explanations
3.2	Borrowing and return UML	Borrowing sequence and return/overdue activity diagrams
4.1	Five-screen UI design	Login, student dashboard, search, details and librarian screens
4.2	Clickable prototype and workflows	Figma prototype with reservation and issue/return workflows
4.3	Usability testing and improvements	Two real classmates' feedback and two evidenced improvements
4.4	Final QA and submission handover	Cross-artifact QA, evidence index and final submission package

3. WBS visual

University LMS | Work Breakdown Structure

16 effort-bearing packages · Talal Ahmad · Phase 1 execution; Assignments 2–4 planned only



Editable sources: reports/wbs.drawio · reports/wbs.csv · reports/planning-data.json. Summary branches and milestones carry no extra effort.

4. Schedule and dependencies

Proposed calendar-day plan, including weekends. Every dependency is finish-to-start: a successor begins after its predecessor ends. Dates are targets, not official deadlines or actual work records. All tasks are assigned to Talal.

ID	Activity	Start → End (2026)	Days	Depends on	Hours
1.1	Initiation and role allocation	13 Nov → 13 Nov	1	None	4
1.2	Repository foundation	14 Nov → 14 Nov	1	1.1	3
1.3	Project charter and scope	14 Nov → 15 Nov	2	1.1	6
1.4	WBS, responsibility and effort	16 Nov → 16 Nov	1	1.3	6
1.5	Schedule and Gantt chart	17 Nov → 17 Nov	1	1.4	5
1.6	Risk assessment and responses	18 Nov → 18 Nov	1	1.5	4
1.7	Issues, commits and A1 evidence review	19 Nov → 20 Nov	2	1.2, 1.5, 1.6	6
2.1	Stakeholder analysis	21 Nov → 23 Nov	3	1.7	7
2.2	Requirements, stories and criteria	24 Nov → 26 Nov	3	2.1	12
2.3	SRS, AI review and traceability	27 Nov → 29 Nov	3	2.2	12
3.1	Use-case and class diagrams	30 Nov → 02 Dec	3	2.3	10
3.2	Borrowing and return UML	03 Dec → 05 Dec	3	3.1	8
4.1	Five-screen UI design	30 Nov → 03 Dec	4	2.3	8
4.2	Clickable prototype and workflows	06 Dec → 08 Dec	3	3.2, 4.1	10
4.3	Usability testing and improvements	09 Dec → 11 Dec	3	4.2	6
4.4	Final QA and submission handover	12 Dec → 14 Dec	3	2.3, 3.2, 4.3	5

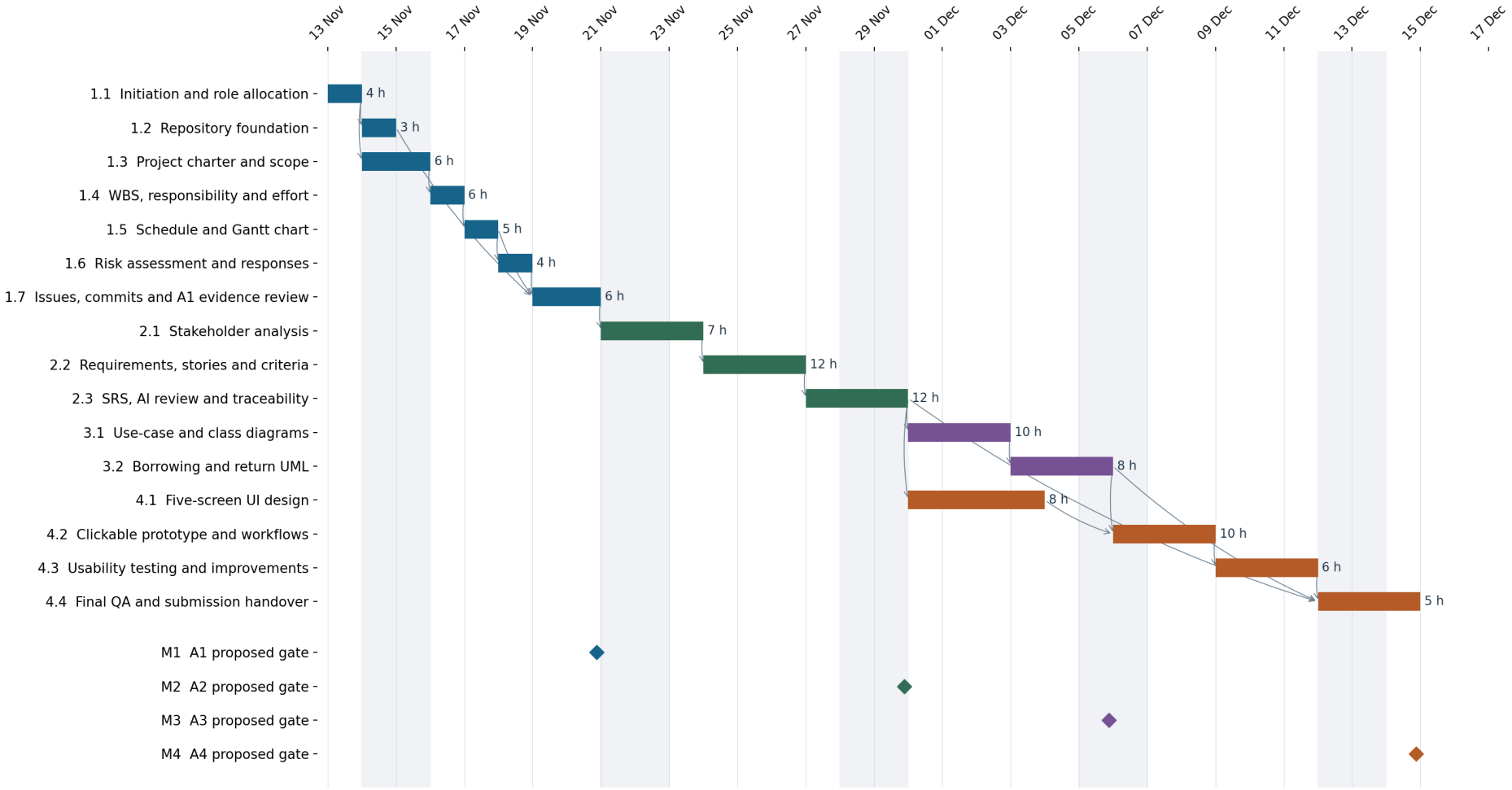
Gate	Proposed target	Official deadline
M1 / A1	2026-11-20	Not supplied
M2 / A2	2026-11-29	Not supplied
M3 / A3	2026-12-05	Not supplied
M4 / A4	2026-12-14	Not supplied

Critical dependency chain: 1.1 → 1.3 → 1.4 → 1.5 → 1.6 → 1.7 → 2.1 → 2.2 → 2.3 → 3.1 → 3.2 → 4.2 → 4.3 → 4.4, 32 calendar days. Task 1.2 has four days of float; task 4.1 has two. Rebaseline when real deadlines or availability are known.

5. Proposed Gantt chart

University LMS | Proposed project schedule

13 November–14 December 2026 • One student: Talal Ahmad • 112 base person-hours



PROPOSED DATES — official deadlines not supplied. Calendar days include weekends. Arrows: finish-to-start; diamonds: end-of-day gates.
A1 blue · A2 green · A3 purple · A4 orange. Later phases are planned only. The schedule is not a record of actual work dates.

6. Responsibility and effort

Talal Ahmad (@talalahmad-ai) is A/R on every row: Accountable for acceptance and Responsible for execution. No additional student is named for Consulted or Informed duties. Functional role allocations describe one person’s types of work, not three people. Estimates are not actual hours or evidence of participation.

ID	Functional lead	Talal RACI	Lead h	Analyst h	Designer h	Talal total h
1.1	Lead	A/R	2	1	1	4
1.2	Lead	A/R	2	0.5	0.5	3.0
1.3	Analyst	A/R	1	4	1	6
1.4	Lead	A/R	4	1	1	6
1.5	Lead	A/R	3	1	1	5
1.6	Designer	A/R	1	1	2	4
1.7	Lead	A/R	4	1	1	6
2.1	Analyst	A/R	1	5	1	7
2.2	Analyst	A/R	2	8	2	12
2.3	Analyst	A/R	2	8	2	12
3.1	Designer	A/R	1	2	7	10
3.2	Designer	A/R	1	1	6	8
4.1	Designer	A/R	1	1	6	8
4.2	Designer	A/R	1	1	8	10
4.3	Designer	A/R	1	1	4	6
4.4	Lead	A/R	3	1	1	5
TOTAL	One student		30	37.5	44.5	112

Phase totals: A1 34 h; A2 31 h; A3 18 h; A4 29 h. Base = 112 h. Unallocated 15% estimate reserve = 16.8 h; envelope = 128.8 h. Reserve is not scheduled task effort or guaranteed calendar float. Combined daily load uses every concurrent task across all roles and does not exceed the assumed six-hour capacity; the overlapping UML/UI tasks peak below that cap.

Review actuals at each gate, reassess any >20% variance and revise dates when capacity is lower or a trigger occurs. sources: effort-estimates.csv, responsibility-matrix.csv and workbook Effort/Capacity sheets.

7. Risks, mitigation and contingency

Likelihood: 1 rare, 2 unlikely, 3 possible, 4 likely, 5 almost certain. Impact: 1 negligible, 2 minor rework, 3 material task delay, 4 phase-gate threat, 5 mandatory submission at risk. Score = L × I. Low 1–5; Medium 6–14; High 15–25. These are qualitative judgments. Talal owns every risk. Review high risks each work session and all risks at every gate.

Highest-priority risks are requirement ambiguity and unknown official deadlines (16 each). Initial repository access problems were resolved and the residual access risk is 10 (2 × 5). The register retains mitigation and contingency actions for every risk.

R01 — Ambiguous lending and reservation policies

Likelihood 4 · Impact 4 · Score 16 · Monitoring role Analyst · Owner Talal Ahmad
Trigger: Conflicting or unanswered policy questions at SRS review
Mitigation: Maintain a policy-question log; prioritize librarian confirmation before baselining.
Contingency: Record explicitly proposed rules, isolate affected requirements and re-review after clarification.

R02 — Effort estimates exceed one student's availability

Likelihood 3 · Impact 4 · Score 12 · Monitoring role Lead · Owner Talal Ahmad
Trigger: Actual hours exceed estimate by 20% or a planned day exceeds six hours
Mitigation: Allocate all role hours to Talal; check aggregate daily demand; review estimates at each gate.
Contingency: Use the 15% effort reserve, re-sequence discretionary polish and propose revised dates to the instructor.

R03 — Repository URL or effective write access fails

Likelihood 2 · Impact 5 · Score 10 · Monitoring role Lead · Owner Talal Ahmad
Trigger: A future push or Issue operation returns an access error; initial HTTP 403 issue is resolved
Mitigation: Use the verified trailing-dot URL; owner connector installation and successful Git push/Issue creation now verify effective access; check again at each gate.
Contingency: Owner restores app access or publishes from an authorized CLI account; retain local commits and package without creating a replacement remote.

R04 — Student unavailable during a critical work package

Likelihood 3 · Impact 4 · Score 12 · Monitoring role Lead · Owner Talal Ahmad
Trigger: Two planned work sessions missed or a milestone is forecast late
Mitigation: Keep tasks small and editable; confirm calendar availability; preserve handover notes.
Contingency: Replan around remaining capacity; request a deadline adjustment if needed; do not invent another contributor.

R05 — Official deadlines differ from proposed dates

Likelihood 4 · Impact 4 · Score 16 · Monitoring role Lead · Owner Talal Ahmad
Trigger: Instructor supplies dates earlier than current proposed gates
Mitigation: Keep every proposed date labeled; request actual submission dates before baselining.
Contingency: Recalculate the dependency network and effort/capacity; obtain revised scope or timing agreement.

R06 — Stakeholders or two classmates unavailable

Likelihood 3 · Impact 4 · Score 12 · Monitoring role Analyst · Owner Talal Ahmad
Trigger: No consultation or testing slot confirmed before the relevant phase
Mitigation: Request availability early and prepare focused questions and a short test script in the appropriate phase.
Contingency: Reschedule with actual eligible participants; disclose missing evidence and seek instructor guidance.

R07 — Git conflicts or accidental loss of existing work

Likelihood 2 · Impact 4 · Score 8 · Monitoring role Lead · Owner Talal Ahmad
Trigger: Remote head changes, conflict appears, or an export is overwritten
Mitigation: Inspect before editing; fetch before pushing; keep editable sources; use normal pushes and reviews.
Contingency: Resolve on a branch, recover from Git and verify against the last reviewed baseline; never force-push.

R08 — Figma access or prototype interactions delayed

Likelihood 3 · Impact 4 · Score 12 · Monitoring role Designer · Owner Talal Ahmad
Trigger: Figma creation, sharing or interaction checks fail at UI kickoff
Mitigation: Confirm account/tool access before design; retain source assets and a workflow checklist.
Contingency: Prepare a local aid and precise Figma completion steps; mark the Figma requirement incomplete until the actual file works.

R09 — SRS, UML and UI use inconsistent rules or terms

Likelihood 3 · Impact 4 · Score 12 · Monitoring role Analyst · Owner Talal Ahmad
Trigger: Review finds a workflow or term that cannot be traced to the baseline
Mitigation: Use a glossary, requirement IDs and cross-artifact review at M2–M4.
Contingency: Pause dependent approvals, fix the source requirement and update all impacted diagrams/screens.

R10 — Testing or screenshots are missing or unreadable

Likelihood 3 · Impact 4 · Score 12 · Monitoring role Designer · Owner Talal Ahmad
Trigger: Evidence index contains a missing file, illegible image or unsupported claim
Mitigation: Capture real pages/artifacts, record URLs/times, open exports and audit every evidence link.
Contingency: Recapture actual state; mark unavailable evidence Pending or Blocked rather than recreating it.

R11 — Scope expands into production implementation

Likelihood 3 · Impact 3 · Score 9 · Monitoring role Lead · Owner Talal Ahmad
Trigger: Tasks add backend, deployment, real data migration or live notifications
Mitigation: Keep the PDF's analysis/design boundary in the charter and review every scope change.
Contingency: Defer implementation requests and restore effort to mandatory assessment artifacts.

R12 — Repository trailing period conflicts with required literal name

Likelihood 3 · Impact 3 · Score 9 · Monitoring role Lead · Owner Talal Ahmad
Trigger: Instructor requires the exact repository spelling from the PDF
Mitigation: Disclose the name difference and ask the owner/instructor to confirm acceptability.
Contingency: Owner renames the existing repository; update origin, document links, Issue map and screenshots; preserve history.

8. Repository management and submission evidence

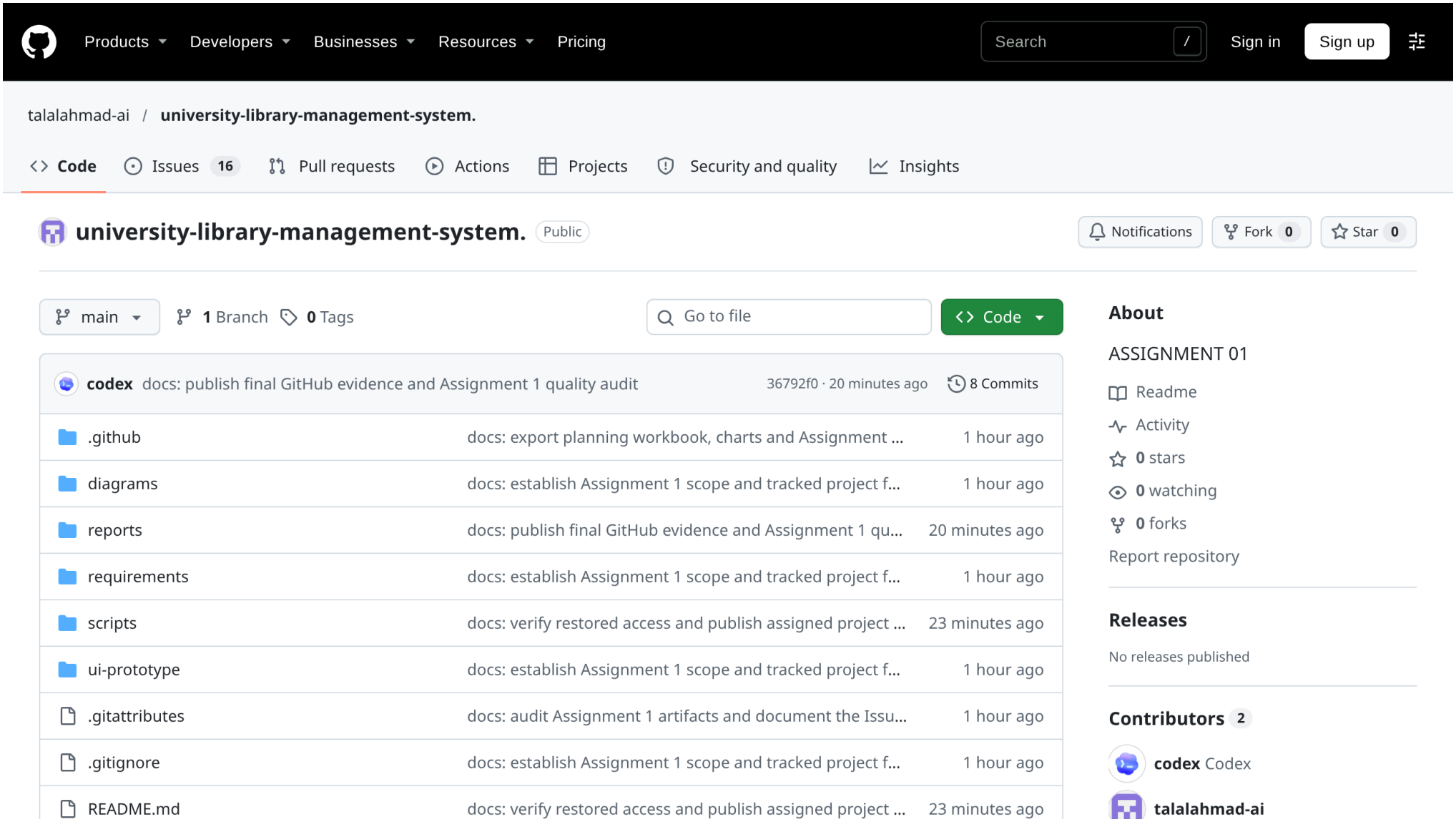
The existing personal repository was cloned and inspected before changes. The original owner commit d6eb076 was retained. Git is available in the execution workspace and the four compulsory folders are tracked. At the verified capture, the remote contained 8 actual commits, including at least three meaningful development commits. Normal non-forced pushes preserve the history.
The 16 major WBS packages each have a real GitHub Issue assigned to talalahmad-ai, with acceptance criteria, phase labels and a matching milestone. Four real milestones use proposed dates in their descriptions; official due dates are unset. The Issues remain open for student review. Future-phase task tracking does not claim execution of Assignments 2–4. Appendices A–F show actual public GitHub pages.

Submission item	Where to inspect
Repository URL / README	Repository URL on cover; complete README in the ZIP; Appendices A–B
Remote commit history	Appendix C; GitHub_Evidence/verification.json in the ZIP
Assigned task Issues	Appendices D–F; live Issues at the repository URL
Gantt / schedule	Planning_Workbook.xlsx; Editable_Planning/; Planning_Charts/
WBS / responsibilities	Editable_Planning/ and Planning_Charts/ in the ZIP
Risk assessment	Section 7; Editable_Planning/risk-register.csv; workbook Risks sheet
Submission checklist	Submission_Checklist.md; verification.json; genuine screenshots in the ZIP

The schedule is an estimate baseline rather than a claim of official deadlines or actual work dates. The existing repository URL includes its literal final period. Talal is the repository owner and the sole named student; all estimated hours belong to him. The assignment does not require a complete LMS implementation.
Source: instructor-supplied Assignements 1 to 4 and Case scenario.pdf, Assignment 1 (page 1). Supporting planning sources and genuine repository evidence accompany this report. Scope: Assignment 1 only.

Appendix A — Repository and required folders

Actual repository homepage showing the tracked project folders and published README.



Source: https://github.com/talalahmad-ai/university-library-management-system. | Captured 12 Nov 2026, 23:50:25 PKT

Appendix B — Project README

Actual GitHub-rendered project name, problem statement, objectives and scope. The complete README is included in the ZIP.

README

University Library Management System

Software Engineering coursework · **Phase 1 / Assignment 1 only** · Reviewed 12 November 2026

Project and problem

The case scenario describes a university library using manual records or an outdated system for membership, book borrowing, returns and overdue items. The proposed Library Management System (LMS) will support books, members, borrowing, returns, reservations and overdue notifications. This coursework specifies and designs that system; the assignment does not require a complete implementation.

Objectives and scope

- Define the project, work packages, responsibilities, effort, dependencies and risks.
- Maintain a transparent GitHub repository with the required folders, meaningful commits and assigned task Issues.
- Plan later requirements/SRS, UML and clickable prototype deliverables as one consistent project.
- Address catalog/copy records, membership, search, issue/return, reservations and overdue notifications at the analysis and design level.

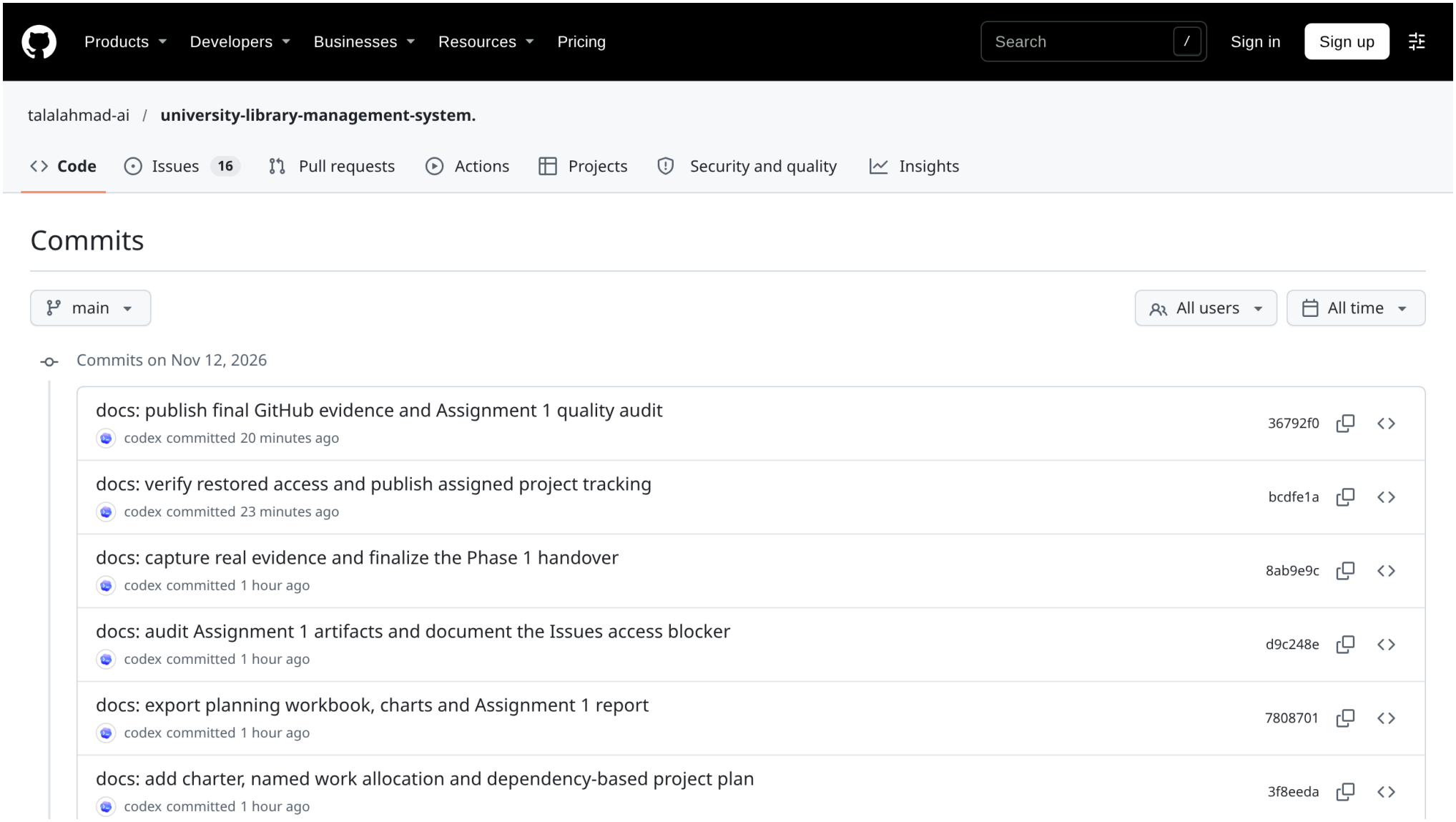
Production software, payment processing, live notifications, deployment, university SSO and real member-data migration are excluded. Borrowing limits, loan periods, fine policies and reservation precedence need stakeholder confirmation. University, department and official deadlines have not been supplied. Only fictional examples may be used.

Stakeholders

Source: https://github.com/talalahmad-ai/university-library-management-system. | Captured 12 Nov 2026, 23:50:26 PKT

Appendix C — Meaningful commit history

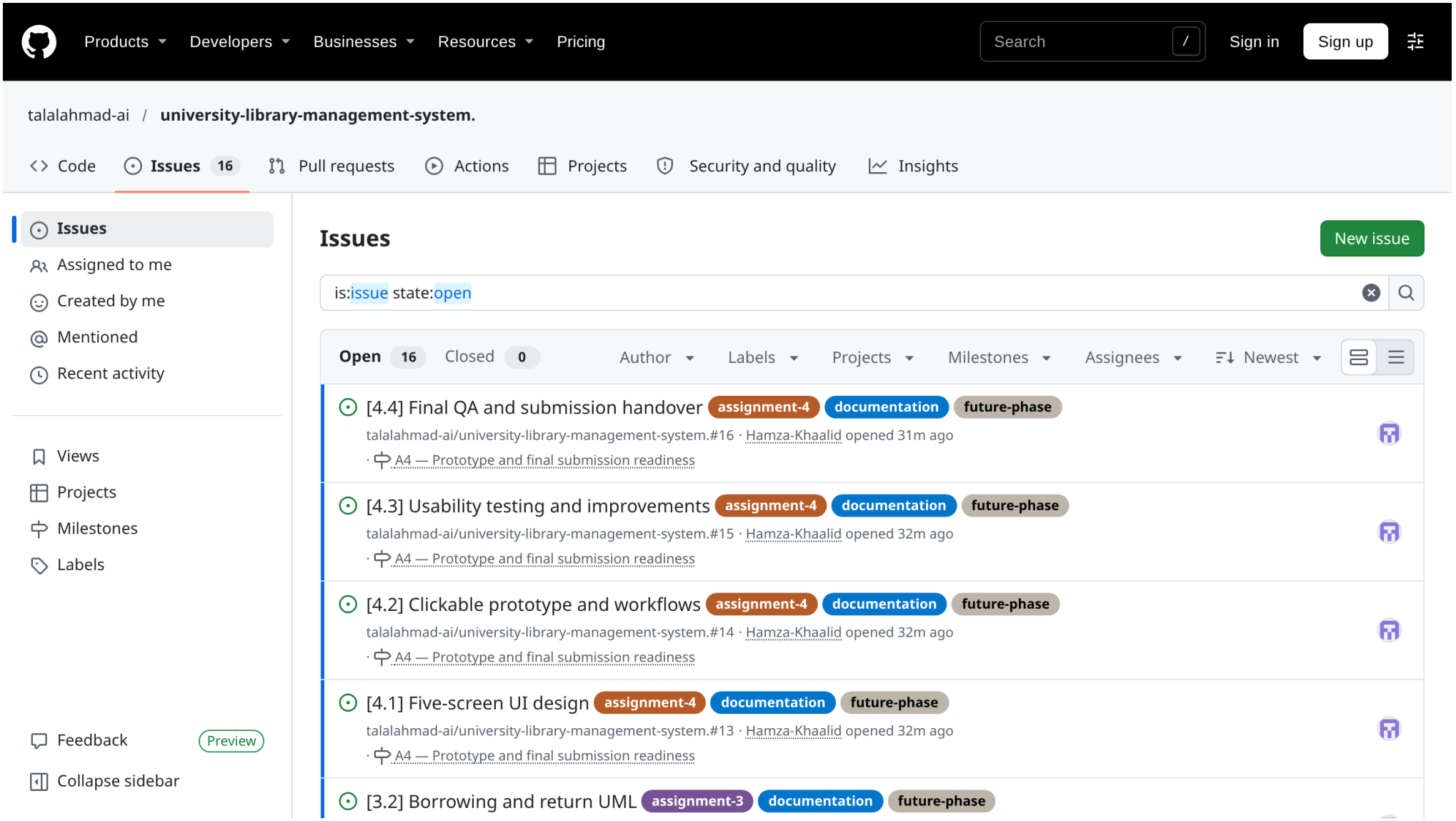
Actual remote development history. The preserved original commit and complete SHA list are recorded in verification.json.



Source: https://github.com/talalahmad-ai/university-library-management-system./commits/main/ | Captured 12 Nov 2026, 23:50:29 PKT

Appendix D — Assigned task Issues

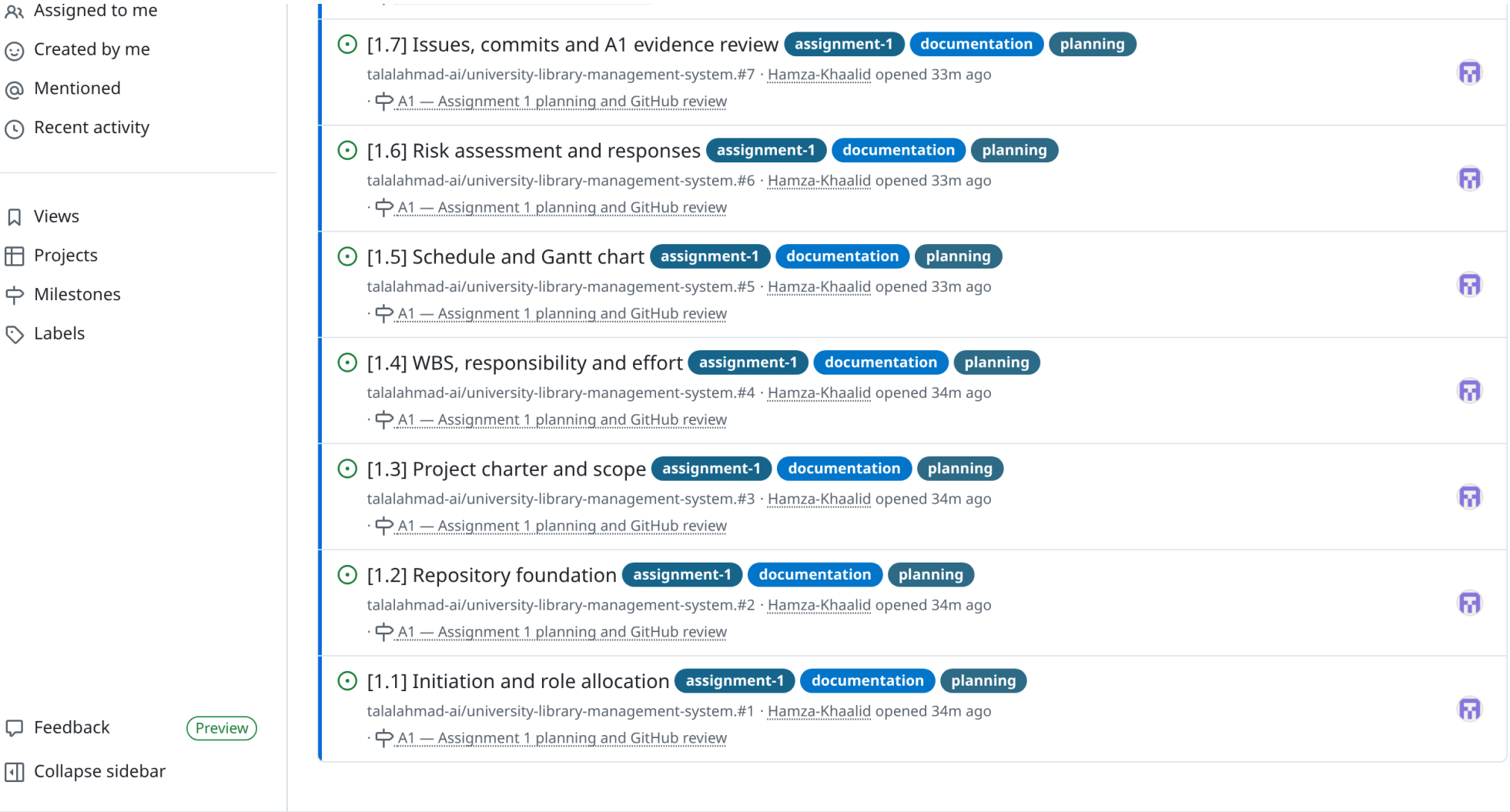
Actual Issues list showing the live task count, phase labels and assignee indicators.



Source: https://github.com/talalahmad-ai/university-library-management-system./issues | Captured 12 Nov 2026, 23:50:31 PKT

Appendix E — Assignment 1 task Issues

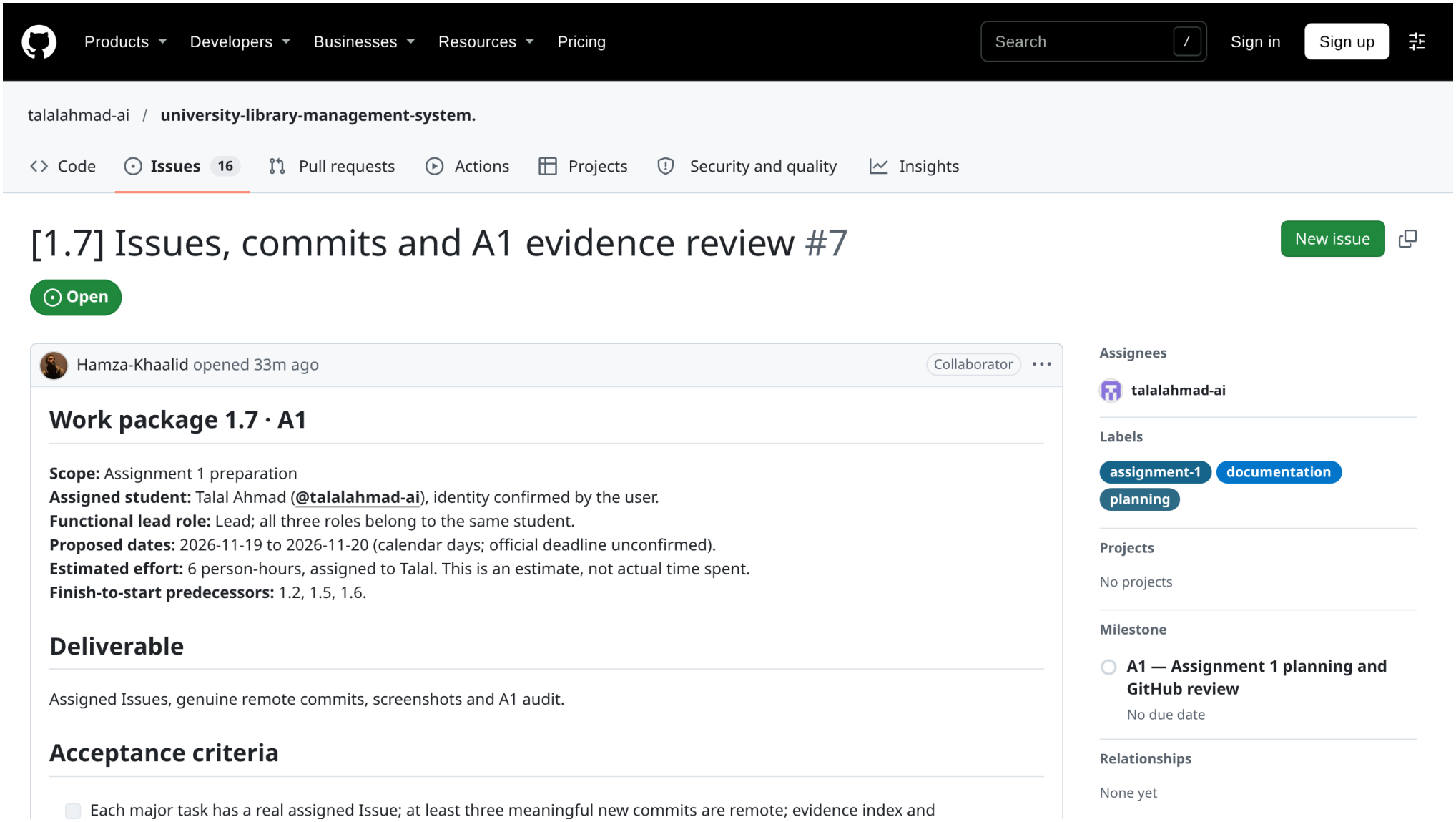
Lower part of the same actual Issues list, including the Assignment 1 planning and GitHub tasks.



Source: https://github.com/talalahmad-ai/university-library-management-system./issues | Captured 12 Nov 2026, 23:50:33 PKT

Appendix F — Individual Issue and assignee

Actual Assignment 1 Issue #7 with its acceptance criteria, talalahmad-ai assignee, labels and phase milestone.



Source: https://github.com/talalahmad-ai/university-library-management-system./issues/7 | Captured 12 Nov 2026, 23:50:35 PKT